VIDYAPEETH

Dropper NEET

KPM

Milestone Test-02

Phase-2

DURATION: 180 Minutes | **DATE: 29/06/2025** | **M.MARKS: 720**

ANSWER KEY

(PHYSICS)

1.	(1)	10.	(2)	19.	(4)	28.	(2)	37.	(1)
2.	(4)	11.	(2)	20.	(3)	29.	(1)	38.	(2)
3.	(4)	12.	(1)	21.	(4)	30.	(1)	39.	(4)
4.	(3)	13.	(1)	22.	(2)	31.	(4)	40.	(2)
5.	(3)	14.	(1)	23.	(2)	32.	(3)	41.	(3)
6.	(4)	15.	(3)	24.	(2)	33.	(2)	42.	(3)
7.	(4)	16.	(2)	25.	(4)	34.	(1)	43.	(3)
8.	(2)	17.	(3)	26.	(3)	35.	(3)	44.	(4)
9.	(3)	18.	(2)	27.	(4)	36.	(4)	45.	(3)

(CHEMISTRY)

46.	(2)	55.	(2)	64.	(4)	73.	(1)	82.	(1)
47.	(1)	56.	(2)	65.	(2)	74.	(2)	83.	(4)
48.	(4)	57.	(2)	66.	(3)	75.	(3)	84.	(3)
49.	(1)	58.	(1)	67.	(2)	76.	(4)	85.	(2)
50.	(1)	59.	(3)	68.	(4)	77.	(2)	86.	(4)
51.	(1)	60.	(4)	69.	(2)	78.	(3)	87.	(2)
52.	(2)	61.	(4)	70.	(4)	79.	(1)	88.	(2)
53.	(2)	62.	(3)	71.	(1)	80.	(2)	89.	(4)
54.	(3)	63.	(4)	72.	(2)	81.	(3)	90.	(3)

(BOTANY)

91.	(1)	100.	(3)	109.	(2)	118.	(4)	127.	(3)
92.	(4)	101.	(3)	110.	(4)	119.	(1)	128.	(3)
93.	(2)	102.	(4)	111.	(1)	120.	(2)	129.	(4)
94.	(2)	103.	(2)	112.	(3)	121.	(3)	130.	(1)
95.	(3)	104.	(1)	113.	(2)	122.	(4)	131.	(3)
96.	(3)	105.	(3)	114.	(1)	123.	(1)	132.	(1)
97.	(1)	106.	(2)	115.	(3)	124.	(4)	133.	(3)
98.	(1)	107.	(1)	116.	(3)	125.	(3)	134.	(3)
99.	(3)	108.	(3)	117.	(4)	126.	(1)	135.	(2)

(ZOOLOGY)

136.	(3)	145.	(4)	154.	(1)	163.	(2)	172.	(4)
137.	(3)	146.	(4)	155.	(2)	164.	(1)	173.	(1)
138.	(3)	147.	(3)	156.	(3)	165.	(4)	174.	(1)
139.	(2)	148.	(2)	157.	(3)	166.	(1)	175.	(1)
140.	(3)	149.	(3)	158.	(1)	167.	(3)	176.	(2)
141.	(3)	150.	(3)	159.	(4)	168.	(2)	177.	(2)
142.	(1)	151.	(2)	160.	(4)	169.	(4)	178.	(3)
143.	(1)	152.	(3)	161.	(4)	170.	(3)	179.	(2)
144.	(1)	153.	(2)	162.	(3)	171.	(1)	180.	(3)

1. **(1)**

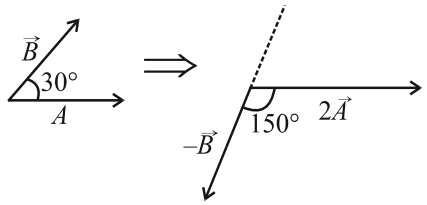
$v_y = u_y - gt$

$= (30 \sin 30°) - (10 \times 1.5) = 0$

Hence, particle is at highest point and angle of velocity with horizontal is zero.

2. **(4)**

When a vector is multiplied by a negative number then the direction of vector will be reversed.



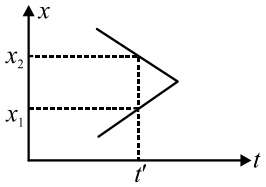
3. **(4)**

$h = \frac{1}{2} g t^2$

$t = \sqrt{\frac{2 \times 20}{10}} = 2 \text{ s}$

$D = ut = 30 \times 2 = 60 \text{ m}.$

4. **(3)**



At a time t', two positions x_1 and x_2 of a particle are not possible.

5. **(3)**

$u = -12 \text{ m/s}$

$s = ut + \frac{1}{2} a t^2$

$= (-12 \times 10) + \frac{1}{2}(-9.8)(10)^2$

$= -120 - 490$

$s = -610 \text{ m}.$

6. **(4)**

$t_1 = \frac{50}{10} = 5 \text{ hr}$

$t_2 = \frac{50}{30} = \frac{5}{3} \text{ hr}$

$t_1 - t_2 = 5 - \frac{5}{3} = \frac{10}{3} \text{ hr}$

= 200 min.

7. **(4)**

Area under $a - t$ graph = change in velocity

$(5 \times 10) + (5 \times 20) = v_{\text{at 10s}} - v_{\text{at 0s}}$

$50 + 100 = v - 5$

$150 + 5 = v$

$v = 155 \text{ m/s}.$

8. **(2)**

Reaction time

$= \frac{\text{Distance vehicle travelled before apply brakes}}{\text{speed of vehicle}}$

$= \frac{10}{20} = 0.5 \text{ s}$

9. **(3)**

Equations of kinematics is valid only for constant Acceleration.

When object fall under the effect of gravity near earth surface, it has

constant acceleration.

Hence, we can use all equations of motion.

10. **(2)**

$T = 10 + 6 = 16 \text{ s}$

$T = \frac{2u}{g} \qquad 16 = \frac{2u}{10}$

$u = 80 \text{ m/s}$

11. **(2)**

Total time of flight = $\frac{2u \sin\theta}{g}$

$\theta_A > \theta_B$, so, $T_B < T_A$

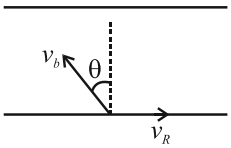
$\therefore$ Shell hits ship B first.

12. **(1)**

Relative acceleration between projectiles is zero. Path of one projectile as seen from another projectile is a straight line.

13. **(1)**

$t = \frac{d}{v_b \cos\theta}$



$\frac{15}{60} = \frac{1}{5\cos\theta}$

$\cos\theta = \frac{4}{5} \Rightarrow \theta = 37°$

$v_b \sin\theta = v_R$

$5 \sin 37° = v_R$

$v_R = 5 \times \frac{3}{5} = 3 \text{ km/hr}.$

14. **(1)**

Direction of angular velocity will not change.

15. **(3)**

$$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 3 & 0\\ 1 & -1 & 1\end{vmatrix}=\hat{i}(3)-\hat{j}(1)+\hat{k}(-1-3)$$

$= 3\hat{i}-\hat{j}-4\hat{k}$

$$\left|\vec{A}\times\vec{B}\right|=\sqrt{9+16+1}=\sqrt{26}$$

$$\hat{n}=\frac{3\hat{i}-\hat{j}-4\hat{k}}{\sqrt{26}}$$

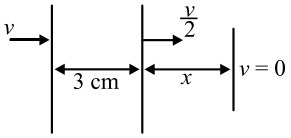
16. **(2)**

$$\omega=\frac{14\times 2\pi}{22}=\frac{28}{22}\times\frac{22}{7}=4\text{ rad/s}$$

$a_c = R\omega^2 = 0.8\times 16$

$= 12.8$ m/s^2.

17. **(3)**



Acceleration is uniform throughout

$$\frac{\frac{v^2}{4}-v^2}{2(3)}=\frac{0-\frac{v^2}{4}}{2(x)}$$

$$\Rightarrow\quad \frac{-3v^2}{4(3)}=\frac{-v^2}{4(x)}$$

$$\Rightarrow\quad x=1\text{ cm}$$

18. **(2)**

For maximum horizontal range, $\theta = 45°$

$\sec 45° = \sqrt{2}$

19. **(4)**

$$\left|\vec{A}\times\vec{B}\right|=|A||B|\sin\theta$$

$= 5\times 10\times\sin 37° = 30$ units.

20. **(3)**

$\left|\vec{v}_{R/G}\right| = \sqrt{(4)^2+(3)^2}$ = 5 km/hr.

21. **(4)**

$$v=\frac{dS}{dt}=12t^2+4t+5$$

$$a=\frac{dv}{dt}=24t+4$$

$$a_{(t=1)}=24+4=28\text{ m/s}^2$$

22. **(2)**

$$a_{avg}=\frac{v_2-v_1}{t_1+t_2}$$

23. **(2)**

$x = 2t^2+3t+1$

$$v=\frac{dx}{dt}=4t+3$$

Velocity depends on time.

$v_{(t=0)} = 4(0)+3 = 3$ m/s

$$a=\frac{dv}{dt}=4\text{ m/s}^2\text{ (constant)}$$

24. **(2)**

$\vec{v}_T = 20\hat{i}$

$\vec{v}_{P/T} = -3\hat{i}$

$\vec{v}_{P/T} = \vec{v}_P - \vec{v}_T$

$-3\hat{i} = \vec{v}_P - 20\hat{i}$

$\vec{v}_P = 17\hat{i}$ (east)

25. **(4)**

$S = -40$ m

$$S=ut+\frac{1}{2}at^2$$

$$-40=u(2)-\frac{1}{2}\times 10(4)$$

$-40 = 2u - 20$

$-20 = 2u$

$u = -10$ m/s.

26. **(3)**

Distance travelled = Area under v-t graph from $t = 0$ to $t = 5$ s.

$$=\frac{1}{2}\times 5\times 4=10\text{ m}.$$

27. **(4)**

Speed at mid-point = $\sqrt{\frac{V_A^2+V_B^2}{2}}$

$$=\sqrt{\frac{(5)^2+(12)^2}{2}}=\frac{13}{\sqrt{2}}\text{ m/s}.$$

28. **(2)**

$$v_P=\frac{d}{dt}(x_P)=10t$$

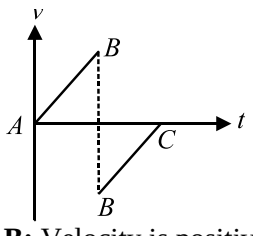
$$v_Q=\frac{d}{dt}(x_Q)=2$$

$10t = 2$

$$t=\frac{1}{5}\text{ s}$$

29. **(1)**

$$v=\frac{dx}{dt}$$



From A to B: Velocity is positive and increasing Hence slope of x-t graph is positive and increasing.

From B to C: Velocity is negative and decreasing, hence slope of x-t graph is negative and its magnitude is decreasing.

30. **(1)**

At $t = 0, v = 0$

$a = 2(t-1)$ so $\frac{dv}{dt} = 2(t-1)$

$\int_0^v dv = \int_0^5 (2t-2)dt$

$v = \left[\frac{2t^2}{2} - 2t\right]_0^5 = (5)^2 - 2(5) = 15 \text{ m/s}$

31. **(4)**

Drunkard walk 2 m forward in 8 second.

So, he cover 8 m forward in 32 second.

After 8 m, he moves 5 steps forward in 5 s and reach to pit and fall into it.

Total time = 32 + 5 = 37 s

32. **(3)**

Speed of a particle can never be negative even if it is moving along $-x$ axis.

33. **(2)**

Time taken by stone to reach the bottom of well

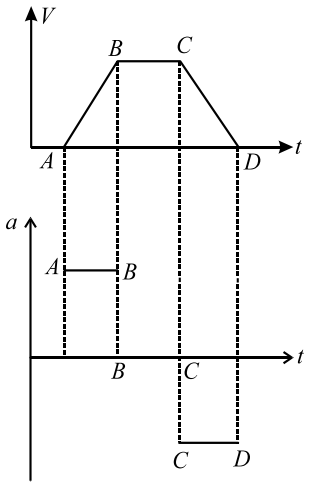
$(t_1)\ t_1 = \sqrt{\frac{2h}{g}}$

Time taken by sound wave to reach the top of well

$(t_2) = \frac{h}{v}$

$T = t_1 + t_2 = \sqrt{\frac{2h}{g}} + \frac{h}{v}$

34. **(1)**



a = slope of $v - t$ graph

A to $B \rightarrow$ slope is positive and constant

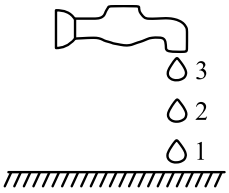
$\Rightarrow$ so acceleration is positive and constant

B to $C \rightarrow$ slope is zero $\Rightarrow$ acceleration is zero.

C to $D \rightarrow$ slope is negative and constant

$\Rightarrow$ so acceleration is negative and constant

35. **(3)**



Time taken by first drop to reach the ground

$\sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 5}{10}} = 1\text{s}$

As water drops are fallings at regular interval from a top therefore time difference between any two drops is 0.5 s

$S = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{2}\times 10 \times (0.5)^2 = 1.25 \text{ m}$

Height from ground = 5 – 1.25 = 3.75 m.

36. **(4)**

For 1st car: $s = 4(t) + \frac{1}{2}(1)t^2 = 4t + \frac{t^2}{2}$ ……(i)

For 2nd car: $s = 2(t) + \frac{1}{2}(2)t^2 = 2t + t^2$ ……(ii)

Equation above equation we get:

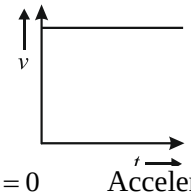
$4t + \frac{t^2}{2} = 2t + t^2$

$\Rightarrow t = 4$ s

Substituting value of t in equation (i);

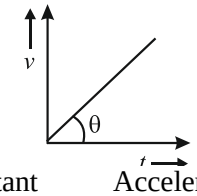
$s = 4(4) + \frac{1}{2}(1)(4)^2 = 16 + 8 = 24 \text{m}.$

37. **(1)**

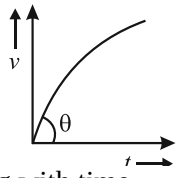


$\tan\theta = \tan 0° = 0$ Acceleration = 0

i.e. v = constant



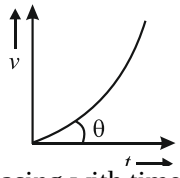
$\tan\theta$ = constant Acceleration = constant



θ is decreasing with time

$\therefore$ $\tan\theta$ is decreasing with time

$\therefore$ acceleration is decreasing with time



θ is increasing with time

$\therefore$ $\tan\theta$ is increasing with time

$\therefore$ acceleration is increasing with time

38. (2)

$|\vec{v}_{M/R}| + |\vec{v}_R| = 16$ ……(i)

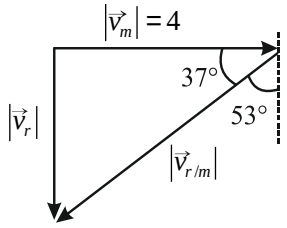
$|\vec{v}_{M/R}| - |\vec{v}_R| = 8$ ……(ii)

Add (i) and (ii)

$2|\vec{v}_{M/R}| = 24$

$|\vec{v}_{M/R}| = 12$ km/hr.

39. (4)



$|\vec{v}_r| = |\vec{v}_m| \tan 37 = 4 \times \frac{3}{4} = 3$ km/hr.

40. (2)

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$ …..(i)

$y = x - \frac{x^2}{2}$ …..(ii)

On comparing (i) and (ii)

$\tan\theta = 1$

$\theta = 45°$

41. (3)

$t = \alpha x^2 + \beta x$ (differentiating with respect to time)

$\frac{dt}{dx} = 2\alpha x + \beta$

$\frac{1}{v} = 2\alpha x + \beta$

Next, we differentiate again with respect to time to find the acceleration:

$-\frac{1}{v^2}\frac{dv}{dt} = 2\alpha\frac{dx}{dt}$

Since $\frac{dx}{dt} = v$. We have

$-\frac{1}{v^2}\frac{dv}{dt} = 2\alpha v$

$\frac{dv}{dt} = -2\alpha v^3$

Therefore, $a = 2\alpha v^3$

42. (3)

$\vec{v}_x = \vec{u}_x = u\hat{i}$

$\vec{v}_y = (u_y - gt)\hat{j} = -gt\hat{j}$

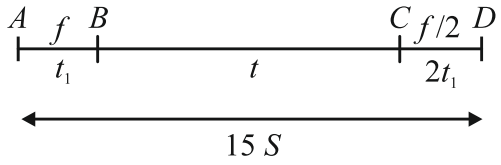
$v_{net} = \sqrt{(u)^2 + (-gt)^2}$

$= \sqrt{u^2 + g^2t^2}$

43. (3)

As, these is no acceleration in the horizontal direction, the magnitude of horizontal component of velocity remains constant.

44. (4)



Initially car starts from rest so $u = 0$

Now distance from A to B.

$S = \frac{1}{2}ft_1^2$

$\Rightarrow ft_1^2 = 2S$

Distance from B to C = $(ft_1)t$

In B to C velocity is constant and $v = ft_1$

Distance from B to C = $(ft_1)t$

In B to C velocity is constant and $v = ft_1$

Distance from C to D

$= \frac{u^2}{2a} = \frac{(ft_1)^2}{2(f/2)} = ft_1^2 = 2S$

$\Rightarrow S + ft_1t + 2S = 15S$

$\Rightarrow ft_1t = 12S$ …(i)

But $\frac{1}{2}ft_1^2 = S$ … (ii)

On dividing the above two equations, we get $t_1 = \frac{1}{6}$

$\Rightarrow S = \frac{1}{2}f\left(\frac{1}{6}\right)^2 = \frac{ft^2}{72}$

45. (3)

$\vec{v}_b = 40\hat{j}$

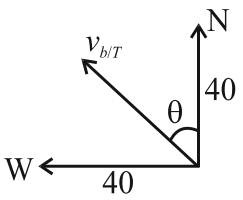
$\vec{v}_T = 40\hat{i}$

$\vec{v}_{b/T} = \vec{v}_b - \vec{v}_T$

$= 40\hat{j} - 40\hat{i}$

$|\vec{v}_{b/T}| = \sqrt{(40)^2 + (-40)^2}$

$= 40\sqrt{2}$ km/hr



$\tan\theta = \frac{40}{40}$

$\theta = 45°$ (North-West)

46. **(2)**

$4Al(s)+3O_2(g)\rightarrow 2Al_2O_3(s)$

Moles of Al = $\frac{4}{3}\times$ moles of O_2

$= \frac{4}{3}\times\frac{1}{2}=\frac{2}{3}$

Weight of Al = $\frac{2}{3}\times 27 = 18\,g$

47. **(1)**

Molecular formula of methane $= CH_4$

Empirical formula of methane $= CH_4$

Molecular formula of ethane $= C_2H_6$

Empirical formula of ethane $= CH_3$

48. **(4)**

(I) $n_{CO_2} = \frac{22}{44} = \frac{1}{2} = 0.5$

(II) $n_{CO_2} = \frac{N}{N_A} \Rightarrow \frac{N_A}{4N_A} = \frac{1}{4} = 0.25$

(III) $n_{CO_2} = \frac{V}{22.4} \Rightarrow \frac{2.24}{22.4} = \frac{1}{10} = 0.1$

49. **(1)**

Moles of urea $= \frac{24.08\times10^{21}}{6.022\times10^{23}}$

Molarity $= \frac{3.99\times10^{-2}}{0.100}$

$= 39.9 \times 10^{-2}$

$= 0.39$

$= 0.4$ M

Hence the concentration of the urea solution is 0.4 M.

50. **(1)**

Zeros at the beginning of a number are not significant.

0.0045 have two significant figures.

51. **(1)**

i. $H_2O \rightarrow$ Molecular mass

$= (2 \times 1) + 16 = 18$ g $\rightarrow$ A

ii. $N_2 \rightarrow$ Molecular mass

$= 2 \times 14 = 28$ g $\rightarrow$ B

iii. $CH_4 \rightarrow$ Molecular mass

$= 12 + (4 \times 1) = 16$ g $\rightarrow$ C

iv. $SO_4 \rightarrow$ Molecular mass

$= 32 + (2 \times 16) = 64$ g $\rightarrow$ D

52. **(2)**

$A_2 + B_2 \rightarrow AB_3$

Applying POAC on A atom,

moles of A atom in A_2 = moles of A atoms in AB_3

moles of A atom in $A_2 = 2$ moles

moles of A atoms in $AB_3 = 1$ moles

2 moles of $A_2 = 1$ mole of AB_3

$\Rightarrow$ moles of $A_2 = \frac{3}{2}$ mole $= 1.5$ mole

53. **(2)**

22.4 L of O_2 contains $= 6.022\times10^{23}$ molecules

1L of O_2 contains $= \frac{6.022\times10^{23}}{22.4}$ molecules

5.6 L of O_2 contains $= \frac{6.022\times10^{23}}{22.4}\times5.6$ molecules

Atoms of O in 5.6 L $= \frac{6.022\times10^{23}\times5.6\times2}{22.4}$

$= 3.011\times10^{23}$ atoms

54. **(3)**

Vapour density $= \frac{\text{Molecular weight}}{16}$

Molecular weight of the substance $= 16 \times 4 = 64$

55. **(2)**

Molality = 3 m = 3 mol/kg

$n_{NaOH} = 3, n_{H_2O} = \frac{1000}{18} = 55.55$

$\chi_{NaOH} = \frac{n_{NaOH}}{n_{NaOH}+n_{H_2O}} = \frac{3}{58.55}$

$\chi_{NaOH} = 0.05$

So, $x_{H_2O} = 0.95$

56. **(2)**

$\frac{n_{H_2}}{3} < \frac{n_{N_2}}{1} \Rightarrow \frac{n_{N_2}}{n_{H_2}} > \frac{1}{3}$

$\frac{W_{N_2}}{W_{H_2}} > \frac{14}{3}$

57. **(2)**

$4Al + 3O_2 \rightarrow 2Al_2O_3$

108 g 96 g 204 g

Mass of O_2 required to react with 1.8 g Al

$= \frac{96\times1.8}{108} = 1.6$ g

$\Rightarrow$ Aluminium is the limiting reagent and O_2 is in excess. Mass of excess

$O_2 = 2.5 - 1.6 = 0.9$ gm

58. **(1)**

Average atomic mass of boron $= \frac{19\times10+81\times11}{100}$

$= 10.8$ u

59. **(3)**

22.4 L of He gas have N_A atoms of He at 1 atm and 273 K

1 mole of any gas occupy 22.4 L at 1 atm and 273 K.

60 **(4)**

$A + 2B \rightarrow C$

According to stoichiometry of reaction, 1 mole of A reacts with 2 mole of B and form 1 mole of C.

So, 5 mole of A will react with 10 mole of B.

So, 2 mole of B produce =1 mole of C

So, 8 mole of B produce = 4 mole of C.

61 **(4)**

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$$

30 20 0

0 5 15

Oxygen left = 20 – 15 = 5 mL

62. **(3)**

(i) All digits are significant except zero in the beginning of a number.

(ii) Zeros to the left of the first non-zero digit are not significant if such zeros follow the decimal point.

(iii) Zeros to the right of the decimal point are significant.

63. **(4)**

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

1 mole of C_3H_8 give 4 mole of H_2O

2 mole of C_3H_8 give 8 mole of H_2O

64. **(4)**

$$\text{Moles of } NO_2 = \frac{2.8}{22.4} = \frac{1}{8}$$

$$\text{Number of oxygen atom} = \frac{1}{8} \times N_A \times 2 = \frac{N_A}{4}$$

65. **(2)**

$$\underset{10\text{ g}}{M_2CO_3} + 2HCl \rightarrow 2MCl + H_2O + \underset{0.2\text{ moles}}{CO_2}$$

Number of moles of M_2CO_3 reacted

= Number of moles of CO_2 evolved

$$\frac{10}{M} = 0.2$$

$$M = \frac{10}{0.2} = 50 \text{ g mol}^{-1}$$

66. **(3)**

In 1 mole of $CaCO_3$, we have:

1 mole of Ca^{2+} ions

1 mole of CO_3^{2-} ions, which contains 1 mole of C atoms and 3 moles of O atoms.

Thus, the total number of atoms = 1 + 1 + 3 = 5 moles of atoms.

Since 1 mole = 6.022×10^{23} entities, the total number of atoms = $5 \times 6.022 \times 10^{23}$.

67. **(2)**

20% (mass/mass) means

100 g of solution has 20 g of KI

$\Rightarrow$ 80 g of solvent contains 20 g of KI

$$\text{Molality, } m = \frac{20}{166} \times \frac{1000}{80} = 1.5 \text{ m}$$

68. **(4)**

Both structures having different molecular formula.

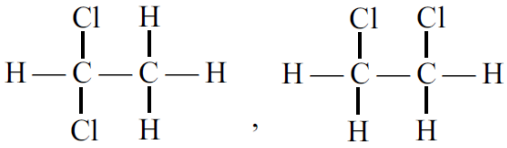
69. **(2)**

$CH_3 - CH_2 - \overset{\overset{O}{\|}}{C} - H$ and $CH_3 - \overset{\overset{O}{\|}}{C} - CH_3$ are example of functional isomerism.

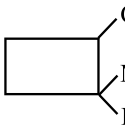
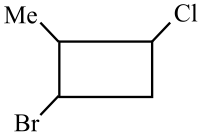
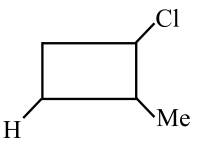
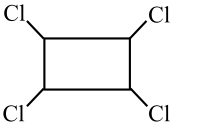
70. **(4)**

Propanal (C_3H_6O) and acetone (C_3H_6O) share the same molecular formula but differ in the position of the carbonyl group (an aldehyde vs. a ketone). Thus they are functional isomers. The other pairs listed do not have the same molecular formula, so they cannot be isomers.

71. **(1)**



72. **(2)**

	(Compound)		(Number of chiral center)
(A)		(III)	1
(B)		(II)	3
(C)		(IV)	2
(D)		(I)	4

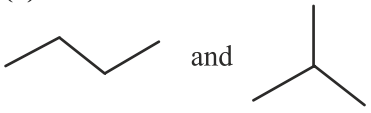
73. **(1)**

Functional isomers possess different functional groups but maintain the same molecular formula. Despite this, their chemical and physical properties can vary significantly due to the diverse functional groups involved.

74. **(2)**

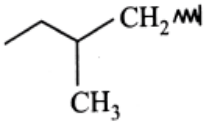
Compounds having the same molecular formula but different functional groups will be functional isomers to each other.

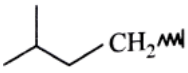
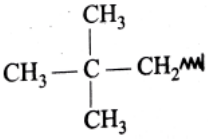
75. **(3)**



76. **(4)**

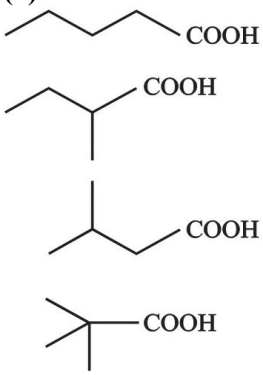
Primary alkyl group means the carbon is attached to parent chain must be essentially CH_2. Hence, with molecular formula (C_5H_{11}) the structures will be-

(i)  (ii) 

(iii)  (iv) 

So, four different primary alkyl groups are possible with C_5H_{11}.

77. **(2)**



78. **(3)**

Compounds having same molecular formula but different alkyl groups attached to the same polyvalent functional group. (like $-O-, -\overset{\overset{O}{\|}}{C}-$) are called metamers of each other.

79. **(1)**

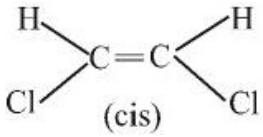
$CH_3–O–CH_2–CH_2–CH_3$ and $CH_3–CH_2–O–CH_2–CH_3$

80. **(2)**

The given molecular formula contains two oxygen atom and hence carboxylic acid and ester functional groups are possible.

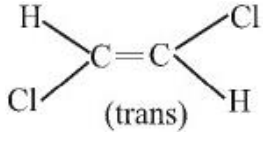
81. **(3)**

The isomer which has similar groups on the same side of the double bonded carbon is called 'cis' isomer



(cis)

The isomer which has similar groups on the opposite side of the double bond is known as 'trans' isomer.



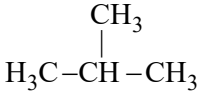
(trans)

82. **(1)**

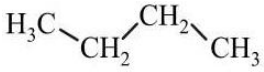
Chain isomerism: It is a type of structural isomerism in which molecular formula is the same but the arrangement of the carbon chain is different. Thus alkanes having more than three carbon atoms exhibit chain isomerism.

For example:- Four carbon atoms:-

One carbon has to be in the middle, two at the ends, but the fourth carbon could be in the middle (forming 2- methylpropane) or at the end (forming butane).



2-methylpropane



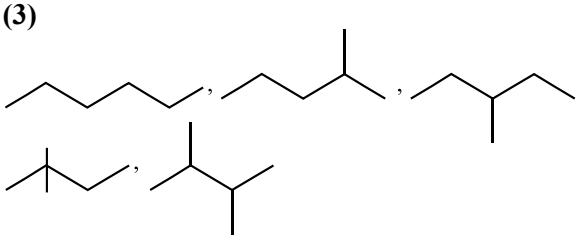
butane

So, minimum four carbon atoms show chain isomerism. Hence, A statement is true. Since every carbon in alkane has 4 sigma bonds, hence, all carbon atoms are hybridised not sp hybridised. Hence, R statement is false.

83. **(4)**

Isomers must have same molecular formula.

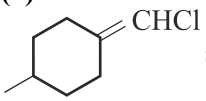
84. **(3)**



85. **(2)**

Total carbon atoms in isomer should be 8 as in given compound total carbon atoms is 8. But in option 2, carbon atoms is 7. So, it can not be isomer of given compound.

86. **(4)**

 shows geomatrical isomerism.

87. **(2)**

Pentanal, 2-methylbutanal, 3-methyl butanal, 2,2-dimethylpropanal, 2-pentanone, 3-pentanone, 2,2-dimethyl-3-pentanone.

88. **(2)**

Positional isomers, which have the same carbon skeleton and functional groups, vary in the positions of the functional groups along or in the carbon chain.

They are positional isomers, hence same molecular formula $C_5H_8Br_2$. However, they are structural isomers, hence different physical and chemical properties.

.

89. **(4)**

For being functional isomers, functional group should not match.

(Phenol and aliphatic alcohol are considered as different functional groups)

90. **(3)**

In option (3) number of carbons are not same.

91. **(1)**
Although cell growth (in terms of cytoplasmic increase) is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle.

92. **(4)**
Growth and reproduction are characteristics of cells, indeed of all living organisms. All cells reproduce by dividing into two, with each parental cell giving rise to two daughter cells each time they divide.

93. **(2)**
A typical eukaryotic cell cycle is illustrated by human cells in culture. These cells divide once in approximately every 24 hours. However, this duration of cell cycle can vary from organism to organism and also from cell type to cell type. Yeast for example, can progress through the cell cycle in only about 90 minutes.

94. **(2)**
The M Phase starts with the nuclear division, corresponding to the separation of daughter chromosomes (karyokinesis) and usually ends with division of cytoplasm (cytokinesis).

95. **(3)**
Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements in telophase of mitosis.

96. **(3)**
The correct sequence of phases in a cell cycle is G1 → S → G2 → M.

97. **(1)**
Prophase is marked by the initiation of condensation of chromosomal material.

98. **(1)**
In an animal cell, cytokinesis is achieved by the appearance of a furrow in the plasma membrane.

99. **(3)**
Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio.

100. **(3)**
The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis, hence the chromosomes are spread through the cytoplasm of the cell.

101. **(3)**
There is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n.

102. **(4)**
A very significant contribution of mitosis is cell repair. The cells of the upper layer of the epidermis, cells of the lining of the gut, and blood cells are being constantly replaced.

103. **(2)**
Although cell growth (in terms of cytoplasmic increase) is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle.

104. **(1)**
Human cells divide once in approximately every 24 hours.

105. **(3)**
Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. In animals, mitotic cell division is only seen in the diploid somatic cells. However, there are few exceptions to this where haploid cells divide by mitosis, for example, male honey bees.

106. **(2)**
Anaphase II begins with the simultaneous splitting of the centromere of each chromosome (which was holding the sister chromatids together), allowing them to move toward opposite poles of the cell by shortening of microtubules attached to kinetochores.

107. **(1)**
The M Phase represents the phase when the actual cell division or mitosis occurs and the interphase represents the phase between two successive M phases.

108. **(3)**
S or synthesis phase marks the period during which DNA synthesis or replication takes place. During this time the amount of DNA per cell doubles.

109. **(2)**
The given figure represents the metaphase stage of the mitosis.

110. **(4)**
Cells in G_0 stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

111. **(1)**
Plant cells are enclosed by a relatively inextensible cell wall, therefore they undergo cytokinesis by a different mechanism.

112. (3)
Yeast can progress through the cell cycle in only about 90 minutes.

113. (2)
In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).

114. (1)
The cell cycle is divided into two basic phases: Interphase and M Phase (Mitosis phase).

115. (3)
The interphase lasts more than 95% of the duration of the cell cycle.

116. (3)
During the division of a cell, DNA replication and cell growth also take place. Cell division is a very important process in all living organisms.

117. (4)
Metaphase of M phase is the stage at which morphology of chromosomes is most easily studied.

118. (4)
M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell.

119. (1)
Mitosis usually results in the production of diploid daughter cells with identical genetic complement.

120. (2)
The interphase, though called the resting phase, is the time during which the cell is preparing for division by undergoing both cell growth and DNA replication in an orderly manner.

121. (3)
The cells that do not divide exit the G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle.

122. (4)
During the Gap 1 phase of interphase, the cell is metabolically active.
Gap 1 phase of interphase corresponds to the interval between mitosis and initiation of DNA replication.

123. (1)
Prophase which is the first stage of karyokinesis of mitosis follows the S and G_2 phases of interphase.

124. (4)
During cell division, golgi complex and ER reform during telophase.

125. (3)
In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).

126. (1)
The formation of the new cell wall begins with the formation of a simple precursor, called the cellplate that represents the middle lamella between the walls of two adjacent cells.

127. (3)
The plants can show mitotic divisions in both haploid and diploid cells.

128. (3)
First stage of karyokinesis is prophase.

129. (4)
By metaphase, condensation of chromosomes is completed and they can be observed clearly under the microscope when the cell is undergoing division.

130. (1)
During mitosis, metaphase chromosome is made up of two sister chromatids, which are held together by the centromere.

131. (3)
Small disc-shaped structures at the surface of the centromeres are called kinetochores.

132. (1)
In animal cells, during the S phase, DNA replication begins in the nucleus, and the centriole duplicates in the cytoplasm.

133. (3)
There is no increase in the chromosome number during S phase. If the cell had diploid or 2n number of chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 2n.

134. (3)
Cells in quiescent stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism.

135. (2)
G in G_1 and G_2 phase of cell cycle stands for 'Gap' where G_1 is also known as Gap 1 and G_2 is known as Gap 2 phase.

136. (3)

A glycosidic bond is formed between two monosaccharide units through a dehydration reaction, leading to the formation of disaccharides and polysaccharides.

137. (3)

Ribose is a five-carbon sugar (pentose) found in RNA and nucleotides like ATP.

138. (3)

Cellulose is a structural polysaccharide found in the cell walls of plants, providing rigidity and strength.

139. (2)

The percentage of proteins in the total cellular mass is 10-15%.

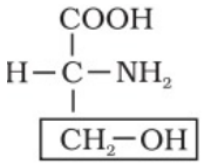
140. (3)

Frogs are poikilotherms as the body temperature of frog varies with temperature of the environment.

141. (3)

Sulphates and phosphates are inorganic compounds that are found in acid-soluble fraction.

142. (1)



Serine

143. (1)

Amino acids are organic compounds containing an amino group and an acidic group as substituents on the same carbon i.e., the α-carbon. Hence, they are called α-amino acids.

144. (1)

A triangular structure called sinus venosus joins the right atrium. It receives blood through the major veins called vena cava.

145. (4)

Inulin is a homopolymer of fructose. Insulin is a polypeptide that works as hormone.

146. (4)

The tympanum (eardrum) helps frogs detect sound vibrations.

147. (3)

Lipids are non-polar molecules and do not dissolve in water, making them hydrophobic.

148. (2)

The molecules in the insoluble fraction with the exception of lipids are polymeric substances.

149. (3)

- Lecithin is a phospholipid.
- A fatty acid has a carboxyl group attached to an R group.

150. (3)

Immunoglobulins (antibodies) help in immune defense by recognizing and neutralizing pathogens.

151. (2)

The secondary structure of proteins involves hydrogen bonding, forming α-helices and β-pleated sheets.

152. (3)

Palmitic acid	16 carbons containing
Glycerol	Trihydroxy propane
Lectins	Concanavalin A
Polymeric substance	Rubber

153. (2)

At the junction of midgut and hindgut is present another ring of 100-150 yellow coloured thin filamentous Malpighian tubules. They help in removal of excretory products from haemolymph.

154. (1)

Glycogen is a storage polysaccharide in animals.Glycogen has a branched structure.

155. (2)

Inulin is a polymer of fructose.

156. (3)

The quaternary structure is formed when multiple polypeptide chains interact (e.g., hemoglobin has four subunits).

157. (3)

In male frogs, a thickened pad develops on the first digit of the forelimb (Label C) during the breeding season. This pad helps males grip the female securely during mating. This is a classic example of sexual dimorphism seen in frogs — a structural modification found in males but not in females.

158. (1)

Frog has different types of sense organs, namely organs of touch (sensory papillae), taste (taste buds), smell (nasal epithelium), vision (eyes) and hearing (tympanum with internal ears). Out of these, eyes and internal ears are well-organised structures and the rest are cellular aggregations around nerve endings.

159. (4)

- Glutamic acid is an acidic amino acid.
- Lysine is a basic amino acid.
- Valine is a neutral amino acid.
- Aromatic amino acids are tyrosine, phenylalanine and tryptophan.
- Serine contains hydroxy methyl group as its functional group.

160. (4)

Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

161. (4)

Collagen is the most abundant protein in animal world and Ribulose bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

162. (3)

Trypsin	Enzyme
Receptor	Sensory reception
Insulin	Hormone
Pigments	Carotenoids

163. (2)

- Tyrosine, phenylalanine, tryptophan are aromatic amino acids.
- Lysine is a basic amino acid.

164. (1)

Phospholipids form the lipid bilayer of cell membranes, with hydrophilic heads and hydrophobic tails.

165. (4)

A triglyceride molecule is made up of one glycerol and three fatty acid molecules.

166. (1)

Chemically modified sugar	Glucosamine
Toxin	Ricin
Drugs	Curcumin
Terpenoides	Monoterpenes

167. (3)

The most abundant chemical in living organisms is water.

168. (2)

The amino acid with a methyl group as its R group is alanine.

169. (4)

The nymphs look very much like adults. The nymph grows by moulting about 13 times to reach the adult form. The next to last nymphal stage has wing pads but only adult cockroaches have wings.

170. (3)

Starch forms helical secondary structures and thus starch can hold I_2 molecules in the helical portion and starch – I_2 complex is blue in colour.

171. (1)

Primary metabolites are formed in animal tissues.

172. (4)

Fatty acids are found esterified with glycerol. They can be then monoglycerides, diglycerides and triglycerides.

173. (1)

Amino acids are substituted methane.

174. (1)

Codeine and morphine are alkaloids.

175. (1)

- Amino acids can be essential or non-essential. essential amino acids have to be obtained through diet.
- Our body can synthesize non-essential amino acids.
- Plant cell walls are made of cellulose.
- Collagen is the most abundant protein in the animal world.

176. (2)

Water > Protein > Nucleic acid > Carbohydrate > Lipids > Ions

177. (2)

In a polypeptide, the long protein chain is folded upon itself like a hollow woolen ball, giving rise to tertiary structure

178. (3)

- Some tissues, especially the neural tissues have lipids with more complex structures.
- Phospholipids have phosphorus and a phosphorylated organic compound present in them. There are found in cell membranes.
- Lecithin is an example of phospholipid.

179. (2)

In a polysaccharide chain (say glycogen), the right end is called the reducing end and the left end is called the non-reducing end.

180. (3)

At physiological pH (~7.4), amino acids exist in a zwitterionic form, where the amino group ($-NH_3^+$) is protonated and the carboxyl group ($-COO^-$) is deprotonated, making the molecule neutral overall but carrying both positive and negative charges.

■■■



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